

Construction and Function of Low-Symmetry Metallo-Supramolecules

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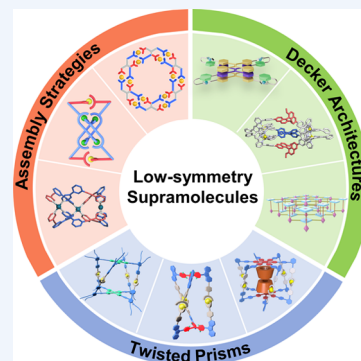
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CONSPECTUS: Natural systems often exhibit high-symmetry structures, which are vital in biological processes. Inspired by nature, chemists have focused on designing high-symmetry metallo-supramolecular architectures due to the high predictability and controllability for self-assembly. In contrast, low-symmetry structures are ubiquitous and also play a key role in living systems. Mimicking natural low-symmetry structures has proven challenging; substantial efforts have been devoted to developing low-symmetry metallo-supramolecules. However, there are still significant challenges in both construction and function of low-symmetry metallo-supramolecules: (1) unexpected structures with similar thermodynamic stability will be generated, resulting in uncontrollable self-assembly; (2) functionalized low-symmetry supramolecular systems are limited by uncontrollable self-assembly. To address the key challenges, we developed various strategies to control the assembly process and explore the functionalities. In this Account, we summarize the recent research progress achieved by our efforts in preparing low-symmetry metallo-supramolecules and exploring their functionalities.

First, the efficient approaches are discussed: (1) precise configurational control by subtly modulating ligands, (2) ligand and metal-ion selectivities enabled by selective self-complementary coordination motifs, and (3) enhancing the steric effect to control coordination modes. Second, the functionality of multidecker structures with well-organized homo/hetero-chromophoric arrangements will be introduced. Finally, a low-symmetry functionalized prismatic system with high universality was developed, which realized a high photoluminescence quantum yield, tunable luminescence, mechanically interlocked structures, and selective encapsulation of low-symmetry guests.



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- Shi, J.; Li, K.; Yu, H.; Han, N.; Yang, T.; Jiang, X.; Hao, X.-Q.; Chen, Z.; Wu, G.; Zhang, H.; Li, B.; Wang, M. Ultra-High Metal-Ions Selectivity Induced by Intramolecular Cation- π Interaction for the One-Pot Synthesis of Precise Heterometallic Architecture. *Angew. Chem. Int. Ed.* **2025**, 64, e202416150.¹ The precise heterometallic architecture is synthesized with ultrahigh metal-ion selectivity in a one-pot method only with terpyridyl motifs.
- Shi, J.; Xu, W.; Yu, H.; Wang, X.; Jin, F.; Zhang, Q.; Zhang, H.; Peng, Q.; Abdurahman, A.; Wang, M. A Highly Luminescent Metallo-Supramolecular Radical Cage. *J. Am. Chem. Soc.* **2023**, 145, 24081–24088.² This work presents the construction of a metallo-supramolecular prism containing a radical species and highly efficient radical luminescence.
- Yu, H.; Li, J.; Shan, C.; Lu, T.; Jiang, X.; Shi, J.; Wojtas, L.; Zhang, H.; Wang, M. Conformational Control of a Metallo-Supramolecular Cage via the Dissymmetrical Modulation of Ligands. *Angew. Chem. Int. Ed.* **2021**, 60, 26523–26527.³ To achieve control of low-symmetry supramolecules, a design strategy to enhance the

dissymmetry of the ligands is proposed, achieving precise configurational control of low-symmetry molecular cages.

- Shi, J.; Li, Y.; Jiang, X.; Yu, H.; Li, J.; Zhang, H.; Trainer, D. J.; Hla, S. W.; Wang, H.; Wang, M.; Li, X. Self-Assembly of Metallo-Supramolecules with Dissymmetrical Ligands and Characterization by Scanning Tunneling Microscopy. *J. Am. Chem. Soc.* **2021**, 143, 1224–1234.⁴ This work shows precise control of coordination modes of metallo-supramolecules with low-symmetry ligands and provides the first visual characterization for low-symmetry metallo-supramolecular system using scanning tunneling microscopy (STM).

1. INTRODUCTION

Natural beauty showcases numerous high-symmetry structures, such as high-symmetry self-assembled protein aggregates,

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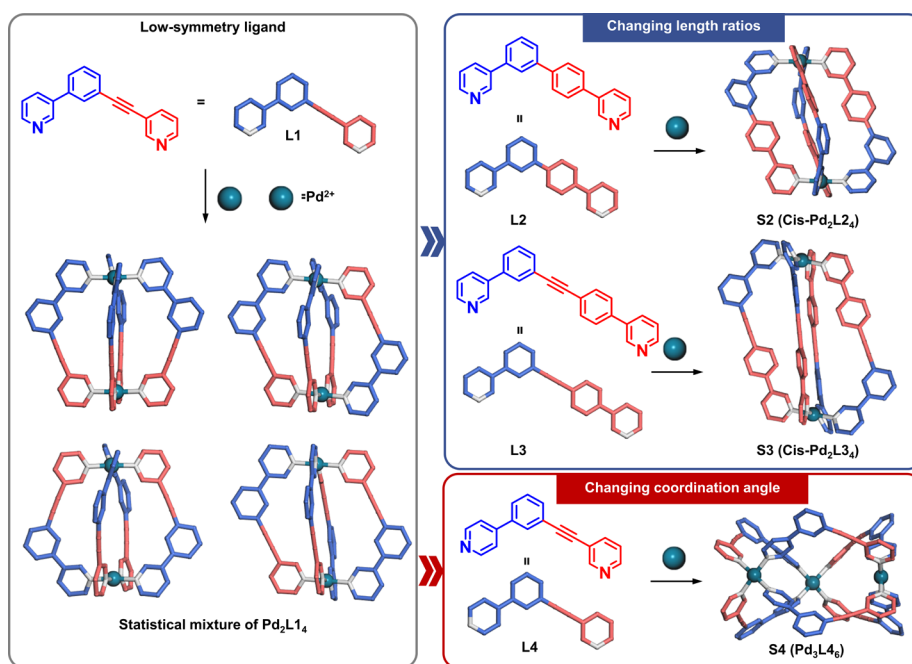


Figure 1. Self-assembly of Pd_2L_{14} , Pd_2L_{24} , Pd_2L_{34} , and Pd_3L_{46} .

coronavirus, DNA, etc., which are vital for a variety of biological processes.⁵ Inspired by those elegant structures, chemists have exploited the self-assembly of labile metal ions and appropriately predesignable ligands to generate high-symmetry metallo-supramolecular architectures.^{6–8} Notably, low-symmetry structures with distinct local microenvironments are also prevalent in nature and play critical roles in biological systems. For example, hemoglobin is composed of four dissymmetry subunits, and the low-symmetry structure ensures oxygen binding, release, and transport regulation.⁹ Likewise, the dissymmetrical arrangement of amino acid residues in the active center of trypsin determines that it can recognize and act on specific peptide bond sequences,¹⁰ thereby ensuring the high catalytic selectivity during the proteolysis reaction. Compared with high-symmetry structures, low-symmetry structures possess fewer symmetry elements (such as points, axes, or planes)^{11,12} and exhibit unique shapes, precise binding sites, and structural diversity.¹³ They display extraordinary sensitivities in recognition, encapsulation, and catalysis, which offer a versatile platform for encapsulation of extremely similar isomers.¹⁴ Constructing low-symmetry structures is a promising candidate for identifying isomers, emulating the uniqueness of biological systems.

Low-symmetry metallo-supramolecules have garnered increasingly attention due to their refined recognition^{14,15} and catalytic ability toward low-symmetry substrates,¹⁶ which push metallo-supramolecular systems to a higher sophisticated level.¹⁷ To mimic natural low-symmetry structures, considerable efforts have been expended on the controllable self-assembly of low-symmetry metallo-supramolecules.^{13,18} However, there are significant challenges in both the construction and function of low-symmetry metallo-supramolecules. First, thermodynamically similar isomers often arise in the assembly process, leading to uncontrolled self-assembly.^{19,20} Second, novel metallo-supramolecular systems are rare, which further restricts the exploration of functionalities.²¹

To address the key challenges for the precise construction and diverse functions of low-symmetry metallo-supramolecules, we have developed various strategies to successfully achieve high

ligand and metal-ion selectivities and enhance material performance. In this Account, we summarized recent research progress achieved by our efforts in preparing low-symmetry metallo-supramolecules and exploring their functionalities. First, we discuss our endeavor in precise formation of low-symmetry supramolecules: (1) precisely controlling the configuration of low-symmetry metallo-supramolecules by subtly modulating ligands, (2) achieving ligand and metal-ion selectivities by developing self-complementary coordination motifs, and (3) enhancing the steric effect to control coordination modes. Second, novel decker architectures have been developed, which achieved well-organized homo/hetero-chromophoric arrangements and significantly enhanced the luminescent performances. Finally, a low-symmetry prismatic system with high universality was developed, which showed outstanding properties, including mechanical interlock, host–guest structure, and excellent luminescence.

2. STRATEGIES FOR PRECISE FORMATION OF LOW-SYMMETRY SUPRAMOLECULES

In low-symmetry supramolecular systems, controllable self-assembly is a primary challenge, since multiple isomers with similar thermodynamic stability could be generated during the self-assembly process, and it is difficult to control the unpredictable orientation of low-symmetry ligands and metal-ion selectivity.^{22,23} Moreover, the characterization of such complex supramolecular arrangements with low-symmetry building blocks is also challenging.^{4,24} In this section, we provide an overview of the strategies that we have developed to achieve a highly selective assembly and visualize characterization.

2.1. Configurational Control of Metallo-Supramolecular Cages via Enhanced Ligand Dissymmetry

To address the issue of isomer coexistence, a more advanced, general, and easily operated strategy was developed via decreasing the symmetry of low-symmetry ligands to construct metallo-supramolecular cages, which allows more complex and

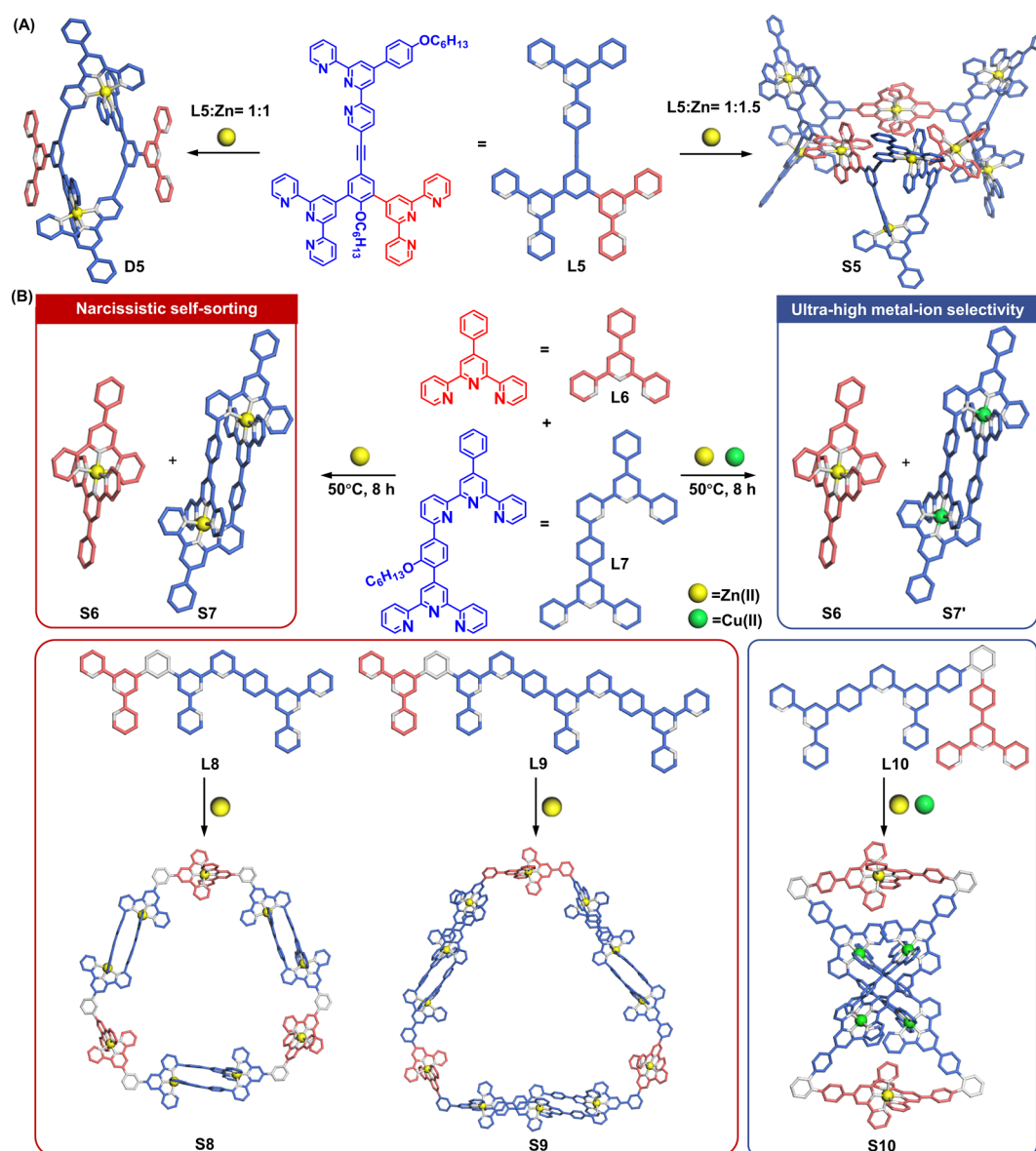


Figure 2. Self-assembly of the hourglass-shaped nanocage **S5** (A), the model metallo-complexes **S6–S7**, the metallo-macrocycles **S8–S9**, and the heterometallic architecture **S10** with Zn^{2+} and Cu^{2+} (B) by the narcissistic self-recognition tpy moieties modified at 5- and 6-positions, respectively.

better configurational control based on low-symmetry building blocks. When ligand **L1** was assembled with Pd^{2+} , multiple isomers of $\text{Pd}_2\text{L1}_4$ were observed, which is attributed to the insufficient low-symmetry feature of **L1** (Figure 1). Compared with **L1**, ligands **L2** and **L3** further increased the length ratios of two arms to enhance dissymmetry and obtain controllable cis-configuration structures ($\text{Pd}_2\text{L2}_4$ and $\text{Pd}_2\text{L3}_4$).³ Encouraged by the successful control of self-assemblies, ligand **L4** was rationally designed by changing the position of the coordination site to further reduce dissymmetry. As expected, the single cis-configuration structure of $\text{Pd}_3\text{L4}_6$ was obtained from nine possible isomeric species, which has been confirmed by nuclear magnetic resonance (NMR) spectrometry, electrospray ionization-mass spectrometry (ESI-MS), and single-crystal X-ray diffraction.

2.2. Developing Selective Self-Complementary Coordination Motifs to Realize High Ligand and Metal-Ion Selectivities

Molecules with complementary features (e.g., donor–acceptor pairs²⁵ and geometry complementarity²⁶) are often employed to design and synthesize complex supramolecular architectures. Among many reported organic ligands, 2,2':6',2''-terpyridine (tpy) has attracted considerable attention due to its ability to coordinate with diverse metal ions and rich coordination sites.²⁷ However, the fixed geometry of tpy-M(II)-tpy connectivity (180°) and poor selectivity, especially metal-ion selectivity, limited the construction of sophisticated architectures.²⁸ To address the above problems, novel tpy motifs with modifications at the 5- and 6-positions were designed, which adopting the head-to-tail coordination mode to give discrete structures and exhibited the narcissistic self-sorting behavior during self-assembly.^{29–31} The hourglass-shaped cage **S5** based on the low-symmetry tritopic ligand **L5** was obtained (Figure 2A).³⁰ To investigate the self-assembly mechanism, **L5** and Zn^{2+} were

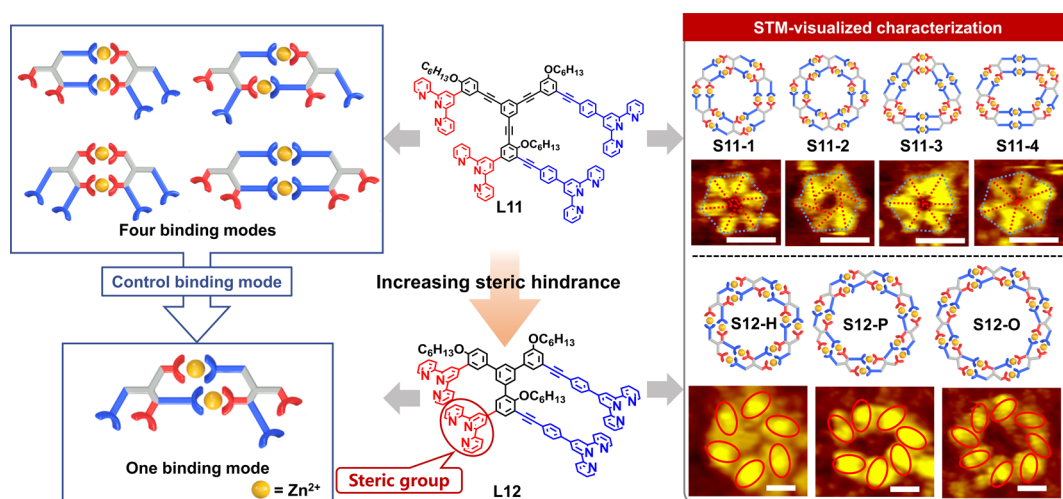


Figure 3. Binding modes of metallo-complexes S11 and S12 and STM images of complexes S11 with a scale bar of 5 nm and S12 with a scale bar of 2 nm.

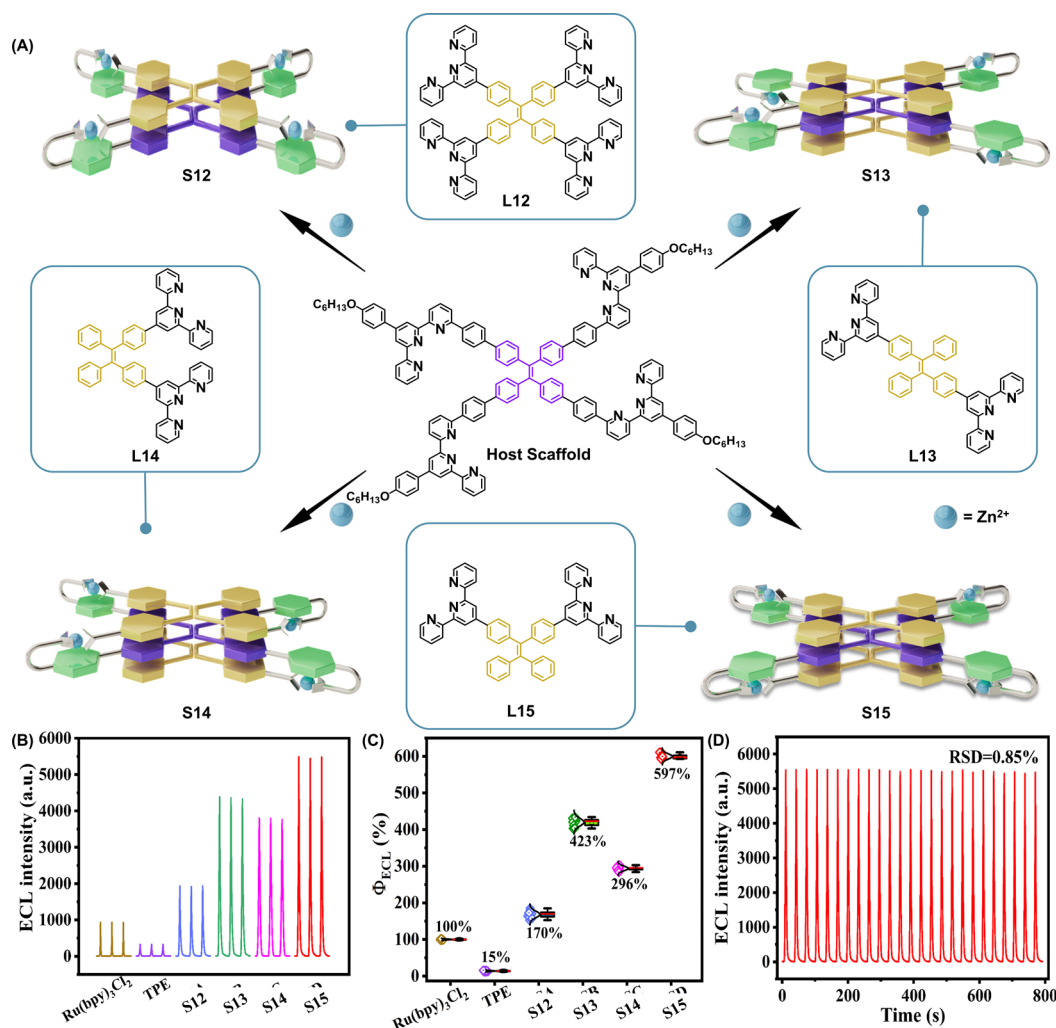


Figure 4. Assembly of multilayer TPE-based architectures S12–S15 (A); the ECL intensity–time curves (B) and ECL efficiencies (C) of 1 mM Ru(bpy)₃Cl₂, TPE, and S12–S15; and ECL intensity–time curve (D) recorded for S15/K₂S₂O₈ system under optimal conditions over 25 continuous cycles.

mixed in the molar ratio of 2:2. ESI-MS spectrum showed only a series of signals of dimer D5, which proved that S5 was obtained based on the excellent selectivity of different tpy units.

For the coordination motif modified at the 6-position, the model system with L6, L7, and Zn²⁺ was first investigated. Only narcissistic self-sorting products S6 and S7 were observed

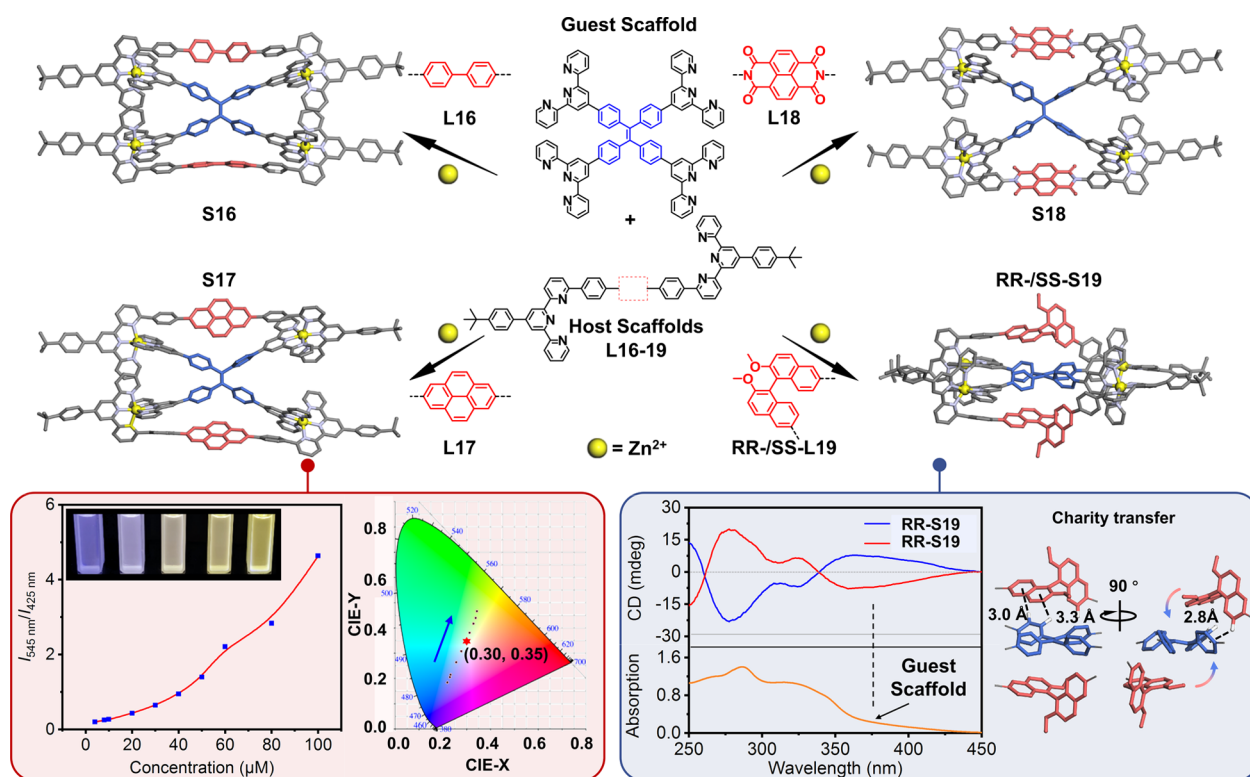


Figure 5. Self-assembly of heterochromophore sandwich-like complexes **S16**–**S19** with guest scaffold and host scaffolds **L16**–**L19** and Zn^{2+} ; the plot and CIE chromaticity coordinates of relative fluorescent intensity ($I_{445\text{nm}}/I_{425\text{nm}}$) of **S17** in different concentration in DMSO with the inset photograph of complex **S17**; CD spectra and absorption spectra of **RR/SS-S19** ($c = 5 \mu\text{M}$) and the C–H... π interactions between binaphthyl and TPE moieties in the X-ray crystal structure of **RR-S19**.

without other mixture (Figure 2B).²⁹ The single crystal of **S7** shows that the coordinated mode is different from that of traditional tpy motifs, and two terpyridyl units in **L7** are noncoaxial and parallel. Based on this excellent selectivity, multitopic **L8** and **L9** with continual dissymmetrical modifications were designed, leading to the formation of hexameric rings **S8** and **S9** without other byproducts or isomers (Figure 2B). In **S7**, the phenyl group between two terpyridyl units of **L7** altered the coordination microenvironment and provided new intramolecular interactions with metal ions. The model system with **L6**, **L7**, Zn^{2+} , and Cu^{2+} showed ultrahigh metal-ion selectivity of Zn^{2+} and Cu^{2+} ; only metallo-complexes **S6** with Zn^{2+} and **S7'** with Cu^{2+} were obtained, respectively.¹ The intramolecular cation- π interactions have been revealed by density functional theory (DFT) calculations. Utilizing this ultrahigh metal-ion selectivity, the heterometallic architecture **S10** based on low-symmetry ligand **L10** was successfully constructed via a one-pot strategy (Figure 2B), featuring a precisely ordered arrangement of two Zn^{2+} and four Cu^{2+} ions. It provided a novel strategy to precisely construct heterometallic architectures.

2.3. Precise Control of Coordination Modes and Visualized Characterization of Low-Symmetry Metallo-Supramolecules

In the field of metallo-supramolecular assembly, beyond the challenge of controllable self-assembly, constructing low-symmetry architectures using building blocks as well as the corresponding characterizations are still in their infancy. In this work, the metallo-supramolecules were assembled with two low-symmetry double-layered ligands and Zn^{2+} (Figure 3).⁴ With the aid of ultrahigh-vacuum, low-temperature scanning tunneling

microscopy (UHV-LT-STM), the four isomeric metallo-supramolecules **S11** with four binding modes were investigated and the statistical data about the population of each isomer (**S11-1**–**S11-4**) were collected at the submolecular level. The distribution of isomers measured by STM was consistent with the sequence of the total binding energy of each isomer obtained by density functional theory (DFT) calculations. To further regulate the coordination mode, the compact ligand **L12** was designed by shortening the linker between two layers; thus, tpy motifs were also as steric blocks. Interestingly, compared with **L11**, only one type of conformation corresponding to a single type of binding mode was found for the self-assembly of **L12**, which is also characterized with UHV-LT-STM. This is the first example of visualized characterization of low-symmetry metallo-supramolecules.

3. FUNCTIONALITIES OF LOW-SYMMETRY DECKER STRUCTURES

In low-symmetry supramolecular systems, the presence of multiple thermodynamically similar isomers severely hinders the precise self-assembly. Therefore, developing effective strategies for building low-symmetry structures is essential not only for precise structural control but also for enabling their functional exploration. Owing to their unique sequences and shapes, low-symmetry structures are capable of recognizing and binding low-symmetry guests, particularly isomers. Precise assemblies are prerequisites for realizing diverse functionalities.

The arrangements of functional groups, corresponding to diverse stacking modes, will affect materials' properties, especially for photofunctional materials.³² Among diverse structures, decker structures have attracted considerable

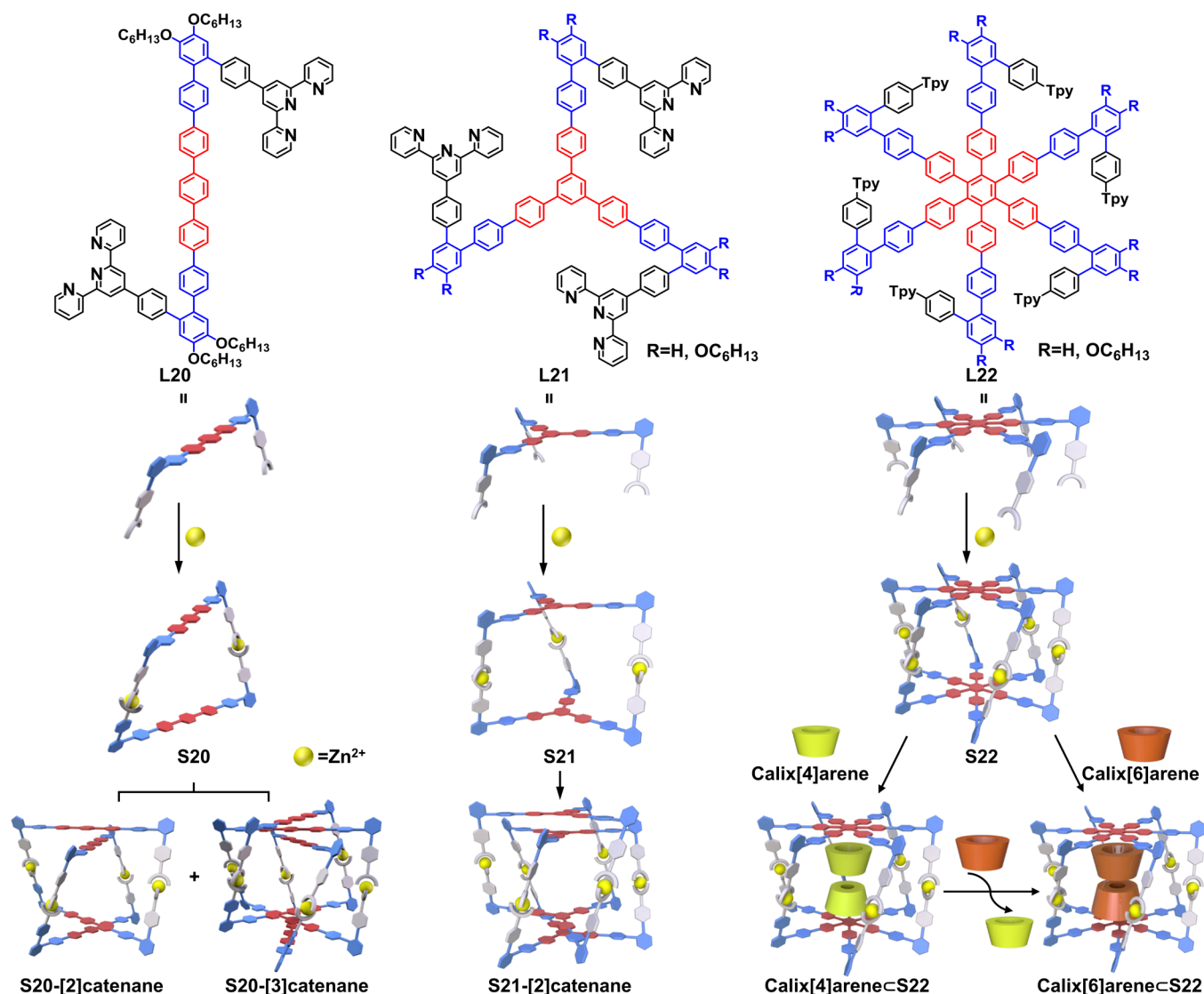


Figure 6. Self-assembly of twisted prisms S20 and S21 with the mechanically interlocked functions, and S22 with the host–guest property.

attention as unique ordered arrays in layer-by-layer structure, which could tune the arrangements of functional groups by molecular design to facilitate the transfer of electrons and exhibit potential application in electronic, photonic, and catalytic fields. In this section, we provide an overview of the low-symmetry metallo-supramolecular decker architectures and their photo-functional properties.

3.1. Novel Multidecker Structures with High Electrochemiluminescent Efficiencies

We reported a series of double- and triple-decker supramolecular architectures, including one double-decker S12 and three “S-shaped” or “X-shaped” triple-decker structures S13–S15, by the shape-complementary coordination strategy through the 6-position modification of tpy and utilization of the abundant angles (60°, 120°, and 180°) of tetraphenylethylene (TPE) (Figure 4A).³³ These decker structures exhibited high electrochemiluminescence (ECL) intensities and efficiencies (Figure 4B–D).³⁴ The best-performing molecule S15 showed a high ECL cathodic efficiency significantly exceeding that of the benchmark tris(2,2'-bipyridyl)ruthenium(II) chloride and TPE in aqueous solution by nearly 6-fold and 40-fold, respectively. The remarkable ECL intensity of these multidecker architec-

tures originates not only from restricted benzene ring rotation and the resulting suppression of nonradiative relaxation but also from the controlled limited stacking of architectures allowing the TPE monomer to be triggered as effectively as possible. Furthermore, triple decker architecture S15 is utilized to realize ECL-based cysteine detection (detection limit = 14.4 nM) in real samples. This study not only provides a precisely modulated supramolecular strategy that enables chromophore regulation at the molecular level but also paves the way for the programmable synthesis of high-performance aggregation-induced electrochemiluminescent emitters. It is a highly efficient and precisely modulated intramolecular aggregation-induced electrochemiluminescent system.

3.2. Multichromophoric Structures with Adjustable Intramolecular Interactions and Chirality Transfer Properties

Multichromophoric nanostructures with well-aligned chromophore arrangements are ideal candidates for investigating interchromophoric interactions and photophysical properties.³⁵ However, developing an effective and customizable strategy to realize effective modulation of the spatial arrangement and interactions of multichromophores still remains a significant

challenge. To further explore the interactions of heteroleptic chromophores, a series of heteroleptic structures **S16–S19** containing different chromophores were designed (Figure 5),³⁶ in which the chromophores are aligned with precise orientations and confirmed by single-crystal X-ray diffraction analysis. Notably, tunable emission and white-light emission were achieved by changing the concentration of **S17**. Encouraged by these results, the strategy was expanded into more complex systems to investigate the functionality. The axially chiral binaphthyl cores were introduced into host ligands to construct sandwich-like structures **RR/SS-S19**. From the single-crystal X-ray diffraction analysis, the conformation of the TPE core was restricted by the C–H... π interactions between TPE and binaphthyl moieties. The circular dichroism (CD) and circularly polarized luminescence (CPL) signals were observed.

4. FUNCTIONS OF TWISTED LOW-SYMMETRY PRISMS

For low-symmetry supramolecular structures, the most important aspects are avoiding complicated molecular design

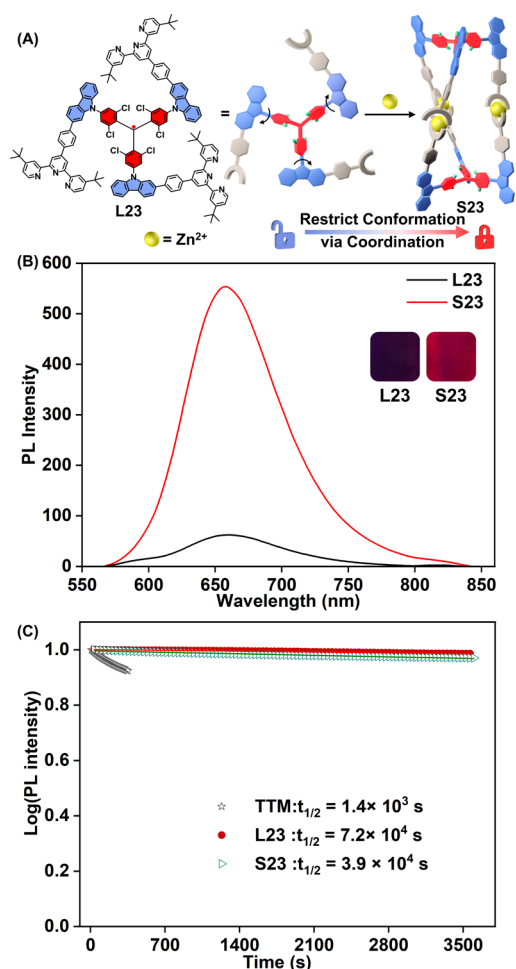


Figure 7. Self-assembly of the trigonal prism **S23** (A), the PL spectra of 1.5 wt % **L23** and 1.5 wt % **S23** doped in PMMA (B) with an excitation wavelength of 375 nm (inset images at 365 nm UV light), and time-resolved fluorescence decay profiles (C) of **L23** and **S23**.

and overcoming the limitations from different numbers of functional group arms to give predictable structures. In this section, based on an innovative design strategy, a series of twisted prisms with well-defined cavities were developed,

demonstrating high universality. In addition, the rigid frameworks of twisted prisms, cooperating with functional groups, exhibited diverse functionalities, including the encapsulation of large guest molecules, formation of mechanically interlocked structures, and efficient and tunable luminescent properties.

4.1. The Twisted Prisms with Interlocked and Host–Guest Properties

Mechanically interlocked molecules (MIMs) and host–guest systems are two compelling areas of supramolecular chemistry that rely on noncovalent interactions to construct complex and functional molecular architectures. A series of ligands (**L20–L22**) with different arms (2, 3, and 6) were designed and coordinated with Zn²⁺ to obtain prisms **S20–S22** (Figure 6).³⁷ As the number of arms increased, space constraints became more pronounced after coordination. For **S20** and **S21**, mechanically interlocked behaviors were observed at higher concentrations, which are confirmed by NMR and ESI-MS. In contrast, **S22** based on the hexatopic ligand has formed a twisted prism without any mechanically interlocked product, due to the substantial space constraints. Single-crystal analysis revealed that the junction angle between the panel and side edges was 63°, resulting in a well-defined and low-symmetry cavity of the twisted prism **S22**. Considering the unique shape and considerable space constraint, calix[*n*]arene (*n* = 4, 6, and 8) was added in excess, respectively. Calix[4]arene and calix[6]arene could be encapsulated in the host/guest ratio of 1:2, but calix[8]arene could not be encapsulated due to its relatively large size and poor solubility. Interestingly, calix[6]arene could be selectively encapsulated when these two types of guest molecules were together added to **S22** and even calix[6]arene was added after prior encapsulation of calix[4]arene. These results illustrated that **S22** exhibited a higher affinity for calix[6]arene than for calix[4]arene.

4.2. The Metallo-Supramolecular Radical Prism with High Luminescence

Based on the successful construction of twisted prisms, the TTM radical core with three arms was introduced. A twisted prism **S23** assembled with tritopic terpyridyl ligand **L23** with tris(2,4,6-trichlorophenyl)methyl (TTM) radical and Zn²⁺ was reported (Figure 7).² The photoluminescent intensity of **S23** was gradually enhanced with the temperature decreasing from 299 to 79 K in 2-methyltetrahydrofuran. Furthermore, **S23** exhibited a remarkable photoluminescence quantum yield (PLQY) of 65%, which is 5 times greater than that of **L23** (13%). The magnetoluminescence of 15 wt % **S23** was observed, while the luminescence of 0.1 wt % **S23** was unaffected by the magnetic field, indicating that the intramolecular spin–spin interaction can be neglected. In addition, **S23** also has excellent photostability (Figure 7C).

4.3. Twisted Prisms with Tunable Fluorescence

Tunable-fluorescence molecules have attracted extensive attention due to their diverse applications in sensing,³⁸ imaging,³⁹ and optoelectronics.⁴⁰ Incorporating chromophores into metallo-supramolecules could disperse the luminescent groups precisely, which is ideal for exploring the tunable luminescence. We introduced the aggregation induced emission (AIE) fluorophore (tetraphenylethylene (TPE)) and aggregation-caused quenching (ACQ) motif (1,3,5-triazine) to verify the universality of prismatic architectures and obtain the prisms **S24** and **S25** (Figure 8A).^{41,42} For **S24** and **S26** without hydrophobic chains, there is no tunable luminescence (Figure

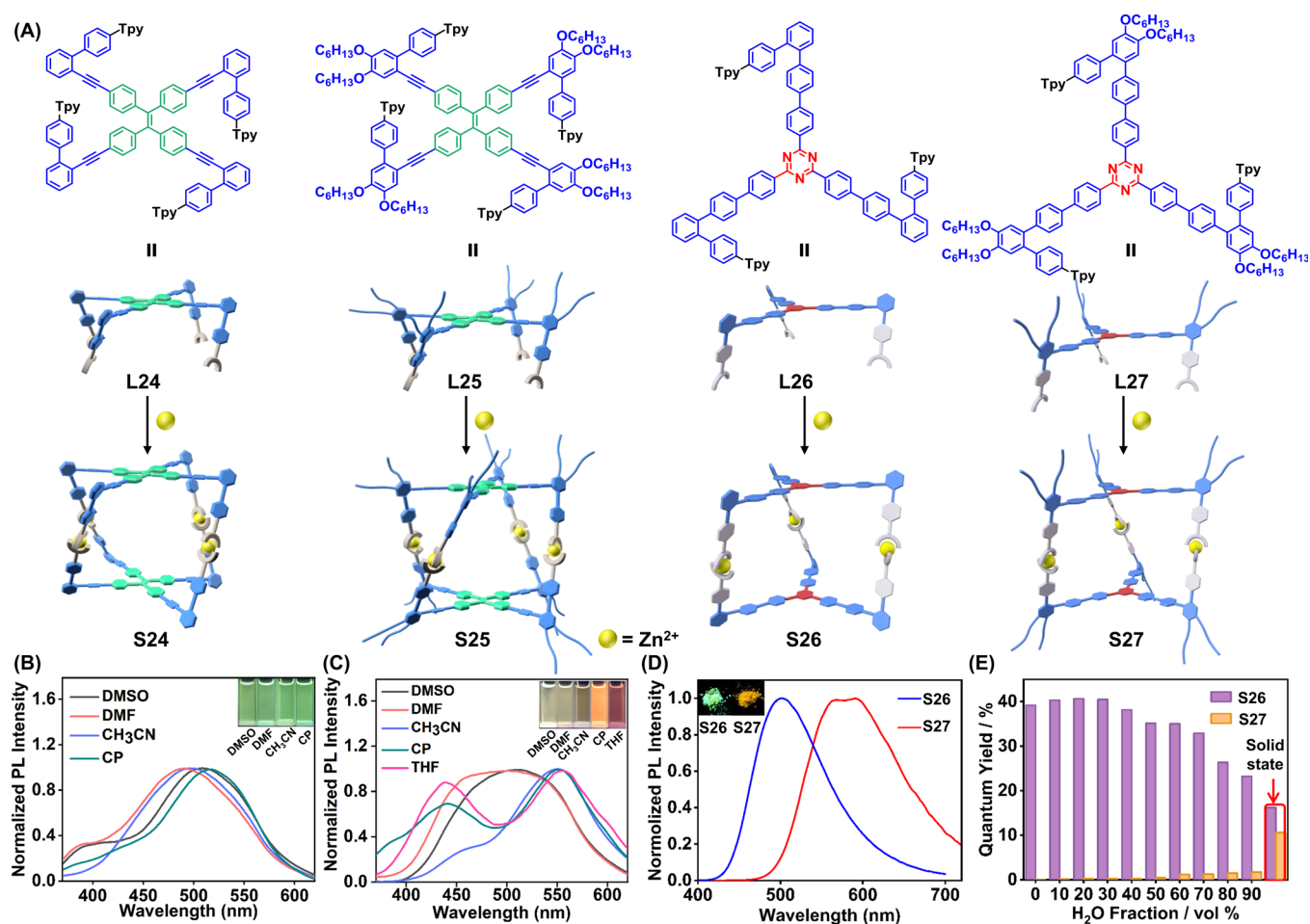


Figure 8. Self-assembly of prisms S24–S27 (A), the solvent effect on the light-emitting properties of S24 (B) and S25 (C) in dilute solution, S26 and S27 in the solid state ($\lambda_{\text{ex}} = 320$ nm) with inset photographs of S26 and S27 at 365 nm UV light (D); quantum yields of S26 and S27 with different fractions of water in the CH₃CN/H₂O solution and solid state (E).

8B) and the hierarchical self-assembly result is spherical aggregates. In contrast, S25 and S27, possessing multiple hydrophobic chains, exhibited tunable luminescent properties in dilute solution (Figure 8C) and formed helical nanofibers through hierarchical self-assembly, which should be attributed to hydrophobic interactions from intermolecular alkyl chains providing the force to direct the molecule stacking in a certain direction. The twisted prisms S26 and S27 have high quantum yields in the solid state (Figure 8D,E), which effectively suppressed fluorescent quenching. These results highlight the effectiveness of twisted prisms as platforms for studying tunable luminescence.

5. CONCLUSION AND OUTLOOK

In summary, effective strategies have been developed for precisely constructing low-symmetry metallo-supramolecules, enabling control over configuration, ligand and metal-ion selectivities, and coordination modes. Additionally, multidecker structures incorporating functional groups have been constructed, exhibiting exceptional properties such as high electrochemiluminescence efficiency, tunable luminescence, and chiral transfer. Moreover, prismatic architectures were developed and demonstrated universality toward different functional units and multifunctionality, including high photoluminescence quantum yield, tunable luminescence, mechanically interlocked structures, and selective encapsulation of low-symmetry guests.

Despite these advances, the current strategies still have limitations in modulating complex self-assembly processes, especially in combining thermodynamic control with dynamic control. Thus, it is crucial to design ligands with different groups to subtly modulate the interactions, further enhancing the controllability of the self-assembly process. Meanwhile, future opportunities and challenges in the field of low-symmetry metallo-supramolecules are highly selective synthesis and functionality of low-symmetry architectures, especially for systems containing multiple metal ions and the synergism of coordination interaction and other noncovalent interactions or covalent bonds. Continued development of low-symmetry structures is expected to unlock novel functionalities and broaden their application potential.

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Author Contributions

The manuscript was written through the contributions of all authors. All authors have given their approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

Biographies

Junjuan Shi completed her Ph.D. in 2021 under the guidance of Professor Ming Wang at Jilin University (China). During 2018–2020, she was a member of the group of Prof. Xiaopeng Li at the University of South Florida (U.S.) in the joint training of doctoral students program. Now she is a Postdoctoral Researcher at Jilin University under the “Dingxin Scholar Program”. Her research interest is the precise construction and functionalities of low-symmetry metallo-supramolecular architectures.

Ming Wang received his BS in 2004 from Jilin University and Ph.D. in 2010 from the Changchun Institute of Applied Chemistry, Chinese Academy of Sciences. After receiving his Ph.D., he joined Professor Pooi See Lee’s group in Nanyang Technological University (Singapore) as a postdoctoral researcher. In 2013, he joined the group of Professor Xiaopeng Li as a postdoctoral researcher at Texas State University (U.S.). In 2016, he was appointed as a professor in the College of Chemistry, Jilin University. His research interests include the construction of metallo-supramolecular architectures, self-assembly, and function based on dissymmetrical ligands.

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